F3XP M.A.R.S.

A Level Physics Practice Paper 1

PHYSICS **9702/41**

Paper 4 A Level Structured Questions **August/September 2026**

MARK SCHEME

Maximum Mark: 100

Practice paper

This mark scheme goes with the practice question paper of the same number. It shows the answers and the marking points that the paper was written to test, set out in the layout of the 2025 Cambridge International mark schemes.

Alternative answers that are correct physics should be credited even where they are not listed.

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This document consists of **10** printed pages.

Mark categories

B marks	Independent marks that do not depend on other marks. The point must be seen in the answer.
M marks	Method marks on which later **A** marks depend. If the **M** mark is not awarded, the **A** marks that follow cannot be awarded either.
C marks	Compensatory marks. They can be awarded even if the point is not written down, when later working shows that it must have been known. If a numerical answer is correct, all the **C** marks before it are awarded automatically.
A marks	Answer marks. They may depend on an **M** mark or allow a **C** mark by implication.

Abbreviations

/	Alternative and acceptable answers for the same marking point.
( )	Bracketed content indicates words which do not need to be explicitly seen to gain credit but which indicate the **context** for an answer.
___	Underlined content must be present in the answer to award the mark.
ecf	Error carried forward: an incorrect earlier answer used correctly later gains the later marks.
allow / ignore	**allow** lists other answers that gain the mark; **ignore** lists content that neither gains nor loses credit.

Question	Answer	Marks
1(a)	direction of motion (and so velocity) changes continuously, so velocity changes **or** velocity is a vector and its direction changes	**B1**
1(b)	horizontal component of tension provides the centripetal force: $T \sin\theta = m\omega^2 r$	**M1**
	$r = L \sin\theta$ so $T \sin\theta = m\omega^2 L \sin\theta$, leading to $T = m\omega^2 L$	**A1**
1(c)	$12 = 0.20 \times \omega^2 \times 0.80$	**C1**
	$\omega = 8.7\ \text{rad s}^{-1}$ The angle θ is not needed: $T = m\omega^2 L$ does not depend on θ.	**A1**
1(d)	horizontal line at $\cos\theta = 1.0$ from $\omega = 0$ to $\omega = 3.5\ \text{rad s}^{-1}$ (string vertical for $\omega \le \sqrt{(g / L)} = 3.5\ \text{rad s}^{-1}$)	**B1**
	for $\omega > 3.5\ \text{rad s}^{-1}$: curve decreasing from $\cos\theta = 1.0$ with decreasing magnitude of gradient ($\cos\theta = mg / T = g / \omega^2 L$)	**B1**
	curve passes through (5.0, 0.49) and (8.0, 0.19), each ± half a small square A line that goes above $\cos\theta = 1.0$ (e.g. through (3.0, 1.36)) cannot score the first mark: $\cos\theta$ cannot exceed 1, so for $\omega < 3.5\ \text{rad s}^{-1}$ the bob does not move in a horizontal circle.	**B1**
1(e)	no change to the values of $\cos\theta$, because $\cos\theta = g / \omega^2 L$ does not depend on m	**B1**
	T doubles, so the string breaks at $\omega = \sqrt{(12 / (0.40 \times 0.80))} = 6.1\ \text{rad s}^{-1}$ and the line now ends at $6.1\ \text{rad s}^{-1}$	**B1**
2(a)	work done per unit mass	**B1**
	(work done) in bringing a small (test) mass from infinity (to the point)	**B1**
2(b)(i)	$GM_E / x^2 = GM_M / (D - x)^2$	**M1**
	$(D - x) / x = \sqrt{(7.35 \times 10^{22} / 5.97 \times 10^{24})} = 0.111$, leading to $x = D / 1.111 = 0.90D$	**A1**

Question	Answer	Marks
2(b)(ii)	curve starting at S, negative throughout, increasing to a maximum (zero gradient) at $x = 0.90D$	**B1**
	curve then decreases to the surface of the Moon, ending at a potential much less negative than at S	**B1**
2(b)(iii)	beyond N the potential decreases towards the Moon **or** beyond N the resultant force is towards the Moon, so the spacecraft accelerates (gains kinetic energy) towards the Moon	**B1**
2(c)(i)	$GMm / r^2 = mv^2 / r$ so $E_K = GMm / 2r$	**C1**
	$\Delta E_K = ½ \times 6.67 \times 10^{-11} \times 5.97 \times 10^{24} \times 450 \times (1 / 6.69 \times 10^6 – 1 / 6.72 \times 10^6)$ $= 6.0 \times 10^7$ J, an increase	**A1**
2(c)(ii)	kinetic energy increases so speed increases: the loss of gravitational potential energy (1.2×10^8 J) is twice the gain in kinetic energy, the other half being the work done against the resistive force	**B1**
3(a)	ratios of temperature need thermodynamic temperatures (the zero of the Celsius scale is not absolute zero): factor is 450 K / 300 K = 1.5	**B1**
3(b)	the vessels are connected so (when the gas settles) the pressure is the same in A and B	**B1**
	gas (molecules) moves from A to B, so the pressure in A rises by less than 50% and the pressure in B increases	**B1**
3(c)(i)	amount of gas is constant: $p_0(V_A + V_B) / T_0 = p(V_A / T_A + V_B / T_B)$	**M1**
	$1.20 \times 10^5 \times 3.0 \times 10^{-3} / 300 = p \times (1.0 \times 10^{-3} / 450 + 2.0 \times 10^{-3} / 300)$ leading to $p = 1.35 \times 10^5$ Pa	**A1**
3(c)(ii)	$pV = NkT$: in A initially $N = 1.20 \times 10^5 \times 1.0 \times 10^{-3} / (1.38 \times 10^{-23} \times 300) = 2.90 \times 10^{22}$ and finally $N = 1.35 \times 10^5 \times 1.0 \times 10^{-3} / (1.38 \times 10^{-23} \times 450) = 2.17 \times 10^{22}$	**C1**
	number = 7.2×10^{21} allow: fraction of the molecules in A falls from 1/3 to 1/4 of the total of 8.70×10^{22}	**A1**

Question	Answer	Marks
3(d)(i)	unchanged, because r.m.s. speed depends only on (thermodynamic) temperature ($\frac{1}{2}m\langle c^2\rangle = 3/2\,kT$); the pressure rises because there are more molecules in B	**B1**
3(d)(ii)	r.m.s. speed □ $\sqrt{T}$	**C1**
	ratio = $\sqrt{(450 / 300)} = 1.2$ Use of Celsius temperatures, $\sqrt{(177 / 27)} = 2.6$, scores 0.	**A1**
4(a)(i)	$\Delta V = 0.150 / 0.598 – 0.150 / 958$ (= 0.251 m^3) and $W = –p\Delta V$	**C1**
	$W = –1.01 \times 10^5 \times 0.251 = –2.5 \times 10^4$ J The negative sign is needed for A1: the steam expands and does work on the atmosphere.	**A1**
4(a)(ii)	$q = mL = 0.150 \times 2.26 \times 10^6 = 3.39 \times 10^5$ J	**C1**
	$\Delta U = q + W = 3.39 \times 10^5 – 2.5 \times 10^4 = 3.1 \times 10^5$ J ecf from (a)(i). 3.6×10^5 J (adding the work done) scores C1 only.	**A1**
4(a)(iii)	temperature is constant so the (mean) kinetic energy of the molecules does not change	**B1**
	(intermolecular) potential energy of the molecules increases as they move apart / bonds are broken	**B1**
4(b)(i)	work done = area enclosed by the cycle = $\frac{1}{2} \times 3.0 \times 10^{-3} \times 3.0 \times 10^5$	**C1**
	= (+)450 J The work done on the gas is positive because the compression (B → C) happens at a higher pressure than the expansion (A → B). –450 J scores C1 only.	**A1**
4(b)(ii)	gas returns to its original state (temperature) so $\Delta U = 0$ for one cycle	**B1**

Question	Answer	Marks
4(b)(ii)	$q = -W = -450$ J, so there is a net transfer of 450 J (of thermal energy) from the gas	**B1**
5(a)	acceleration (directly) proportional to displacement (from a fixed point)	**B1**
	acceleration in the opposite direction to displacement **or** acceleration (always) directed towards the fixed point	**B1**
5(b)(i)	$\omega = 2\pi / 0.80$ (= 7.85 rad s^{-1})	**C1**
	$v_0 = \omega x_0 = 7.85 \times 0.040 = 0.31$ m s^{-1}	**A1**
5(b)(ii)	$v = \omega\sqrt{(x_0^2 - x^2)} = 7.85 \times \sqrt{(0.040^2 - 0.020^2)}$	**C1**
	$v = 0.27$ m s^{-1}	**C1**
	fraction = $(0.272 / 0.314)^2 = 0.75$ allow $E_K / E = (x_0^2 - x^2) / x_0^2 = 3/4$. An answer of 0.5 scores 0.	**A1**
5(c)	at the equilibrium position the tension in the spring equals the weight ($ke = mg$), so only the equilibrium extension is smaller on the Moon	**B1**
	for a displacement x from equilibrium the resultant force is $-kx$, which does not depend on g	**B1**
	$a = -(k / m)x$, so $\omega^2 = k / m$ and the period is unchanged	**B1**
6(a)	charge / potential allow charge per unit potential	**B1**
6(b)	$V = Q / 4\pi\varepsilon_0 r$ and $C = Q / V$ leading to $C = 4\pi\varepsilon_0 r$	**M1**
	$C = 0.10 / 8.99 \times 10^9 = 1.1 \times 10^{-11}$ F	**A1**

Question	Answer	Marks
6(c)(i)	charge is conserved: $Q = 1.11 \times 10^{-11} \times 40 \times 10^{3} = 4.4 \times 10^{-7}$ C and $C_Y = 0.050 / 8.99 \times 10^{9} = 5.6 \times 10^{-12}$ F	**C1**
	$V = Q / (C_X + C_Y) = 4.4 \times 10^{-7} / 1.67 \times 10^{-11} = 27$ kV allow $V = 40 \times 0.10 / 0.15 = 27$ kV	**A1**
6(c)(ii)	energy = ½QV **or** ½CV^2: initially 8.9×10^{-3} J, finally 5.9×10^{-3} J	**C1**
	energy dissipated = 3.0×10^{-3} J One third of the initial energy is dissipated, whatever the resistance of the wire.	**A1**
6(c)(iii)	converted to thermal energy in the wire (by the current in its resistance) allow emitted as electromagnetic radiation	**B1**
6(c)(iv)	$E = Q / 4\pi\varepsilon_0 r^2 = V / r$, with V the same for both spheres	**C1**
	ratio = 0.10 / 0.050 = 2.0 Y has half the charge of X but a quarter of the surface area, so the field at the surface of the smaller sphere is the greater.	**A1**
7(a)	the constant / direct voltage	**M1**
	that produces the same (mean) power in a resistor (as the alternating voltage)	**A1**
7(b)(i)	$P = V^2 / R$: $12^2 / 18 = 8.0$ W and $6.0^2 / 18 = 2.0$ W	**C1**
	mean power = (8.0 + 2.0) / 2 = 5.0 W Use of the mean p.d. (3.0 V) or of 12 / √2 scores 0.	**A1**

Question	Answer	Marks
7(b)(ii)	$V_{r.m.s.} = \sqrt{(5.0 \times 18)} = 9.5$ V allow $\sqrt{90}$ V; ecf from (b)(i)	**A1**
7(c)(i)	$\tau = RC = 1.0 \times 10^3 \times 100 \times 10^{-6} = 0.10$ s and $t = 1 / 50 = 0.020$ s	**C1**
	$V = 12\ e^{-0.020\ /\ 0.10} = 9.8$ V	**C1**
	ripple voltage = 12 – 9.8 = 2.2 V	**A1**
7(c)(ii)	capacitor now discharges for (about) half the time (0.010 s) between maximum values	**B1**
	discharge time is much less than RC, so the decrease in p.d. is approximately proportional to time Exact values: 1.14 V for full-wave and 2.18 V for half-wave rectification.	**B1**
8(a)	electric force (qE) and magnetic force (B_1qv) act in opposite directions	**B1**
	no deflection only when the forces are equal: $qE = B_1qv$, so $v = E / B_1$; ions with other speeds are deflected (and hit the barrier)	**B1**
8(b)(i)	$v = E / B_1 = (1.5 \times 10^3 / 0.015) / 0.50 = 2.0 \times 10^5$ m s^{-1}	**C1**
	$r = mv / Bq = 20 \times 1.66 \times 10^{-27} \times 2.0 \times 10^5 / (0.40 \times 1.60 \times 10^{-19}) = 0.104$ m (and 0.114 m for neon-22)	**C1**
	distance = 2 × (0.1141 – 0.1038) = 0.021 m The ions travel in semicircles, so the separation is the difference in diameters. 0.010 m scores C1 C1 only.	**A1**
8(b)(ii)	speed selected is unchanged (2.0×10^5 m s^{-1}) because both forces double / $v = E / B_1$ does not depend on charge	**C1**
	radius = 0.114 / 2 = 0.057 m, so distance = 0.11 m	**A1**

Question	Answer	Marks
8(c)	(magnetic) flux linking each ring changes as it falls, so (by Faraday's law) an e.m.f. is induced in **both** rings	**B1**
	the complete ring is a closed circuit so there is an induced current in it; there is no current in the ring with the slit	**B1**
	the current produces a (magnetic) force that opposes the motion (Lenz's law), so the acceleration is less than g; no current means no force on the ring with the slit	**B1**
9(a)(i)	Planck constant = gradient of the line	**C1**
	e.g. $6.0 \times 10^{-19} / ((14.0 - 5.0) \times 10^{14}) = 6.7 \times 10^{-34}$ J s allow 6.5×10^{-34} to 6.8×10^{-34} J s	**A1**
9(a)(ii)	work function energy = $hf_0 = 6.7 \times 10^{-34} \times 5.0 \times 10^{14} = 3.3 \times 10^{-19}$ J allow from the intercept of the extrapolated line on the $E_{K\ max}$ axis	**A1**
9(b)	straight line parallel to the line for metal X	**B1**
	line meets the f-axis at 7.5×10^{14} Hz threshold frequency = $5.0 \times 10^{-19} / 6.7 \times 10^{-34}$; allow 7.4 to 7.6 $\times 10^{14}$ Hz	**B1**
9(c)	no change, because $E_{K\ max}$ depends only on the photon energy (frequency); more photons per second only increase the rate of emission of electrons	**B1**
9(d)	$\lambda = h / p$ and $p = \sqrt{(2mE_K)}$, with the same E_K	**C1**
	ratio = $\sqrt{(m_p / m_e)} = \sqrt{(1.67 \times 10^{-27} / 9.11 \times 10^{-31})} = 43$	**A1**
9(e)	binding energies: ^{2}H 2 × 1.11 = 2.22 MeV, ^{3}H 3 × 2.83 = 8.49 MeV, ^{4}He 4 × 7.07 = 28.28 MeV	**C1**

Question	Answer	Marks
9(e)	energy released = 28.28 – (2.22 + 8.49) = 17.6 MeV The neutron has no binding energy. 7.07 – 1.11 – 2.83 = 3.13 MeV scores 0.	**A1**
10(a)	Z of air is very different from Z of skin / soft tissue, so (almost) all of the ultrasound would be reflected at an air–skin boundary	**B1**
	Z of gel is close to Z of soft tissue, so (almost) all of the ultrasound is transmitted into the body	**B1**
10(b)	$I_R / I_0 = (7.80 – 1.63)^2 / (7.80 + 1.63)^2 = 0.43$	**C1**
	total path in tissue = 12 cm: $e^{-0.23 \times 12} = 0.063$	**C1**
	ratio = 0.43 × 0.063 = 0.027 Use of a path of 6.0 cm (giving 0.11) scores max 2.	**A1**
10(c)	λ_{max} = 320 nm for P and 960 nm for Q; $\lambda_{max}T$ = constant, so $T_P / T_Q = 3.0$	**C1**
	$L = 4\pi\sigma r^2T^4$ so $L_Q / L_P = 45^2 \times (1 / 3.0)^4$	**C1**
	ratio = 25 The cooler star Q is 25 times as luminous as P because of its much larger radius.	**A1**
10(d)	$v = c\Delta\lambda / \lambda = 3.00 \times 10^8 \times (678.9 – 656.3) / 656.3 = 1.03 \times 10^7$ m s^{-1}	**C1**
	$t = d / v = 4.6 \times 10^{24} / 1.03 \times 10^7 = 4.5 \times 10^{17}$ s = 1.4×10^{10} years $t = d / v = 1 / H_0$, so any galaxy that obeys Hubble's law gives the same estimate.	**A1**